FAMILY HANDBOOK

Wiggins Survival Book

Staying warm, fed, and safe on the eastern plains of Colorado
For our family — Morgan County, Colorado

Edition: October 2026

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General guidance for one family, not a substitute for a doctor, licensed electrician, certified chimney sweep, or official emergency instructions. Call 911 in an emergency.

CHAPTER 1

How to Use This Book & the First 72 Hours

Who this book is for

This is the survival handbook for **our family in Wiggins, Colorado** (Morgan County, eastern plains, about 4,500 feet). It is written for:

- **Daniel** and his **wife**
- **The kids:** a newborn (breastfeeding, born late September 2026), a 1-year-old (on formula), a 2-year-old (on regular milk), and a 6-year-old
- **Grandma and Grandpa**, next door

We have **chickens** (about 6–12 eggs a day), **many rabbits**, a **propane tank** feeding a modern gas furnace and central AC, and an **old wood stove** in the basement that is not hooked up yet but **we do have a chimney**.

Our biggest local threats are **blizzards and deep cold**, **long power outages**, **high wind**, **tornadoes and hail**, **grass fires**, **summer heat and drought**, and **flooding along the South Platte**.

How to read it

- Read it **once, all the way through, now** while times are normal. Most of what keeps a family alive happens **before** an emergency.
- Each chapter ends with a **"Do it now" checklist**. Check things off in pencil.
- **The Home Worksheet** (in the phone app's Appendix — it is not in the printed book) is where we write down our shutoffs, propane, well, chimney, medicines, and contacts. Fill it in and keep it updated.
- Keep **one printed copy** in the kitchen and **one at Grandma and Grandpa's**. Put it in a gallon zip bag so it survives water.
- Phones die. Paper doesn't. If the power is out, use the paper copy.

> **Rule of thumb:** This book is a guide, not a doctor, electrician, or chimney sweep. **Use professionals now, while they are available.** When in doubt about a baby's health, call the doctor or 911.

The rule of threes (what kills first)

People usually die in this order when things go bad:

1. **Air / carbon monoxide** — minutes. *(Generators, grills, or heaters used indoors.)*

2. **Cold (exposure)** — hours in a Colorado blizzard. Babies and older adults lose heat fastest.
3. **Water** — about 3 days. Babies dehydrate much faster.
4. **Food** — weeks.

So the priority list is always: **safe air → warmth → water → food → everything else.**

Survival mindset (from the U.S. Army Survival Manual)

The Army's survival manual (FM 21-76, now ATP 3-50.21) builds everything on the word **SURVIVAL**:

- **S**ize up the situation, yourself, and your gear.
- **U**se all your senses — and **U**ndue haste makes waste. Don't rush into mistakes.
- **R**emember where you are (and where help, water, and fuel are).
- **V**anquish fear and panic.
- **I**mprovise.
- **V**alue living — stubbornness to get the family through matters.
- **A**ct like the natives — learn how locals (old ranchers, Grandma and Grandpa) handled it.
- **L**ive by your wits, but for now, **learn basic skills.**

When something goes wrong, use **STOP: Stop, Think, Observe, Plan** — before acting.

Lessons from people who lived it

- **Texas, February 2021 (Winter Storm Uri):** millions lost power for days in deep cold. Families learned that **no power means no heat and often no water**, pipes burst, and **diesel fuel can gel in very cold weather** (keep anti-gel additive in diesel trucks and generators). Many died of **hypothermia and carbon monoxide** from improvised heating. *(The Prepared; CDC)*
- **The 1888 "Schoolhouse Blizzard" on the Great Plains** hit on a mild day with almost no warning; many people, including children walking home from school, died caught outside. **Lesson: when a plains blizzard hits, stay put in shelter.**
- **Eastern Colorado, December 2006:** two blizzards struck about a week apart. See Chapter 12.

The first hour

1. **Get everyone together** (both houses). Count heads: 8 people.
2. **Check for danger:** gas smell, fire, injuries, downed power lines. If you smell gas: **get out, don't flip switches, call 911 / the propane company from outside.**
3. **Turn on the battery NOAA Weather Radio** (Chapter 10) to learn what is happening.
4. **Fill water containers and bathtubs now** while water pressure still works (Chapter 2).

5. **Decide: stay or go?** Most of the time we **shelter in place**. Leave if: fire is approaching, floodwater is rising, the house is unsafe, or officials order evacuation.

The first 24 hours

- **Pick one warm room** if heat is out (Chapter 3). Close off the rest of the house.
- **Keep the fridge and freezer shut.** A full freezer holds food safely about 48 hours, a half-full one about 24 hours, if the door stays closed. A fridge holds about 4 hours. *(USDA)*
- **Check on babies every hour:** warm? feeding? wet diapers?
- **Ration phone battery.** Text instead of calling. Turn on low-power mode.
- **Check animals:** water for chickens and rabbits. In winter, it freezes fast.
- **Write down what happened and when** on the log page at the back.

The first 72 hours

- **Make a supply count** (Chapter 13 inventory sheets): water, formula, diapers, food, propane %, firewood, medicines.
- **Set a routine:** meals, baby feeds, chores, sleep shifts, someone on watch for fire/CO.
- **Combine households if heat or water is short.** Two houses heated is twice the fuel. Usually it is better for everyone to live in **one** house.
- **Start conserving immediately.** It is easier to ease up later than to run out.

If we must leave (evacuation)

Keep a **go-bag** for each person, ready by the door (see Chapter 13). Pick meeting places:

- **Spot 1:** End of our driveway (fire in the house).
- **Spot 2:** A place in town (fill in on the worksheet).
- **Spot 3:** Out of the area — a relative or friend (fill in).

Leave a note on the door saying where we went and when.

Do it now

- ☐ Read this whole book once.
- ☐ Fill in the Home Worksheet.
- ☐ Print two copies; one at each house.
- ☐ Buy a battery/hand-crank **NOAA Weather Radio**.
- ☐ Put **battery CO alarms** on every level of both houses.
- ☐ Sign up for **Morgan County emergency alerts** (ask the County Sheriff's Office / Office of Emergency Management) and turn on Wireless Emergency Alerts on all phones.

CHAPTER 2

Water

You can live weeks without food but only **about 3 days without water** — and babies, nursing moms, and older adults much less. Water is the first thing to secure.

How much water we need

The standard rule (FEMA / Ready.gov): **at least 1 gallon per person per day** for drinking and basic hygiene. Store **at least a 2-week supply** if you can.

Nursing mothers, sick people, and anyone in hot weather need **more**. Plan extra.

Who	Gallons/day (minimum)	2 weeks
7 people eating/drinking (all but newborn)	7	98
Extra for nursing mom, formula, cooking, heat	+2	+28
Family total	**~9**	**~126 gallons**
Chickens & rabbits	see Chapters 5–6	

Our goal: about 125+ gallons stored for people, plus water for animals.

Storing water

- Use **food-grade** containers: store-bought bottled water, 5–7 gallon water jugs, or 55-gallon food-grade barrels (blue).
- **Do not use** milk jugs (they break down and leak) or containers that held chemicals.
- Store in a cool, dark place, off concrete (put on a pallet or wood).
- **Replace stored tap water every 6 months.** Unopened commercial bottled water: follow the date. *(Ready.gov)*
- **Don't let it freeze and split** — keep inside the heated part of the house in winter, or leave room for expansion.
- Hidden water in the house: the **water heater tank** (often 40–50 gallons), pipes, and toilet tanks (**not** bowls; and not if you use tank cleaners).

Draining the water heater in an emergency

1. **Turn off the gas/power to the water heater first.**
2. Close the cold-water inlet valve on top.
3. Open a hot faucet upstairs to let air in.
4. Drain from the valve at the bottom into a clean container. The first water may have sediment — let it settle.

Making water safe

If the water might be contaminated (well flooded, town "boil water" notice, river or stock tank water):

1. Boil (best)

- Let cloudy water settle, then filter it through a clean cloth or coffee filter.
- Bring to a **rolling boil for 1 minute**. *(Above 6,500 ft boil 3 minutes; Wiggins is below that.) (CDC)*
- Let cool. Pour between clean containers to improve taste.

2. Bleach (when you can't boil)

Use **plain, unscented household bleach** (no "splash-less" or scented). Check the label for the percent.

Bleach strength	Per 1 gallon of clear water	Per 5 gallons
6%	**8 drops** (about 1/8 tsp)	40 drops (½ tsp)
8.25%	**6 drops**	30 drops (about ⅓ tsp)

- **Double the dose if the water is cloudy or very cold.**
- Stir and **wait 30 minutes.** It should smell slightly of chlorine. If not, repeat and wait another 15 minutes. *(EPA)*
- Bleach loses strength over time — replace yearly.

3. Filters

A good gravity or squeeze filter (such as those rated to remove bacteria and protozoa) is very useful, but **most filters do NOT remove viruses or chemicals.** For the safest water: **filter, then boil or bleach.**

What none of these fix

Chemicals, fuel, fertilizer, or pesticides (common after floods in farm country) are **not** removed by boiling or bleach. If water smells of fuel or chemicals, **don't use it.**

Water for the babies

- **The newborn needs no water** — only breast milk. Never give a young baby plain water.
- **Formula water:** use **bottled water** or water that has been **boiled**. If unsure of water safety, use **ready-to-feed formula** (needs no water at all) — this is the best emergency choice. See Chapter 4.
- Wash bottles in safe water; boil bottles and nipples if you can't wash properly.

If we have a private well

Most wells use an **electric submersible pump. No power = no water.** Plan one of these **now**:

1. **Generator + transfer switch or interlock** installed by a licensed electrician so the well pump can run. Well pumps often need a large generator — check the pump's label (volts/horsepower) and ask the electrician.
2. **Manual well hand pump** installed alongside the electric pump. Several brands make pumps that share the well casing. **Whether this works depends on our static water level** — deep water levels need special (and more expensive) pumps. Get our well log (worksheet) and ask a licensed well contractor.
3. **Solar well pump** (costly, but works for years).
4. **Stored water** (above) to bridge the gap.

After a flood: a flooded well may be contaminated. Don't drink it until it is disinfected (shock chlorinated) and tested. Colorado labs and CSU Extension give guidance; the county health department can help.

Freezing: keep the well house/pressure tank area above freezing — insulate, use a heat lamp **only if** fire-safe, and keep the pump area dry.

If we have town water

- Town water usually keeps flowing for a while in outages, but pressure may drop and **boil-water notices** may come. Listen to the radio.
- **Fill everything when trouble starts.**
- Know where the main shutoff is — if a pipe breaks, shut it off fast.

Winter: keeping pipes from freezing

- Keep the house above freezing. Open cabinet doors under sinks on outside walls.
- Let a faucet **drip** on very cold nights (if water supply allows).
- **If we lose heat for days:** shut off the main water, open all faucets (top floor first, then lowest), and flush toilets to drain lines. Put a little RV antifreeze in drains and toilet bowls if you have it. *(Do not drink RV antifreeze.)*
- **Snow is water:** melt it — but treat/boil it like any untreated water, and it takes a lot of snow (and fuel) to make a little water. Never eat snow to stay hydrated — it chills the body.

Sanitation (just as important as water)

Dirty hands and human waste spread diarrhea, which can kill babies quickly through dehydration.

- **Wash hands** with soap and water after the toilet, diapers, animals, and before food. Hand sanitizer (60%+ alcohol) if water is short — but it doesn't work well on dirty hands.
- **If toilets can't flush:** use a **bucket toilet**: a 5-gallon bucket lined with two heavy trash bags. Add kitty litter, sawdust, or wood ash after each use. Seal and store bags outside away from water and animals. Keep a lid on.
- **Don't** put human waste or used diapers in the chicken or garden compost.
- Keep the bucket toilet **away from the kitchen and water storage.**
- Dish water: wash dishes in hot soapy water, rinse in water with a little bleach (about 1 tablespoon bleach per gallon of water), air-dry.

River and stock water

The **South Platte** and ditches near farm fields can carry farm runoff and animal waste. Only use them if you must — **settle, filter, then boil.** Never let kids play in floodwater.

Do it now

- [ ] Store ~125 gallons of drinking water (start with cases of bottled water and a few 5–7 gal jugs).
- [ ] Buy unscented bleach and an eyedropper; tape the dose chart to the jug.
- [ ] Buy a water filter (bacteria + protozoa) and a large stockpot for boiling.
- [ ] Learn how to drain our water heater.
- [ ] **Well:** get the well log; talk to an electrician (generator inlet) and a well contractor (hand pump).
- [ ] Bucket toilet kit: 2 buckets with lids, heavy bags, kitty litter, TP, gloves.
- [ ] Mark the main water shutoff with a tag.

CHAPTER 3

Heat & Shelter in Winter

On the eastern plains, a winter storm with wind can drop wind chills far below zero. **Without heat, a house can become dangerously cold within a day or two.** Babies and Grandma and Grandpa are at highest risk. This chapter covers keeping warm with propane, with the wood stove, and with neither.

> ⚠ **Carbon monoxide (CO) is one of the biggest killers in winter power outages.** It has no smell. Every year families die from generators, grills, and heaters used indoors. **Put battery-powered CO alarms on every level, and near every sleeping area, in both houses — now.**

Our heat sources, in order

1. **Propane furnace** (normal) — needs **electricity** to run.
2. **Wood stove** (basement) — reconnected to our existing chimney, **after professional inspection**.
3. **Indoor-rated propane heater** (backup, used exactly per the label).
4. **Body heat, layers, and one insulated room** (always works).

1. The propane furnace

No power = no furnace

A modern gas furnace uses electricity for the **igniter, controls, and blower fan**. When the power is out, **it will not run** even with a full propane tank.

Fix (do this now, if we choose a generator): have a **licensed electrician** install a **generator inlet with a transfer switch or interlock**, so a generator outside can power the furnace (and well pump, fridge, lights). **Never "backfeed" a generator through a dryer plug** — it can kill line workers and burn the house.

Making propane last

- One gallon of propane holds about **91,500 BTU** of heat.
- Read the tank gauge (percent full). Gallons left ≈ tank size × gauge % ÷ 100. Example: 500-gal tank at 60% ≈ 300 gallons.
- **Write the reading daily** in the log (Chapter 13). After 3 days we'll know our burn rate: *gallons used per day → days left.*
- Conserve:
- Set the thermostat lower (around 60–62°F) in the house, and keep the **babies' room warmer**.

- Close vents and doors to unused rooms.
- Cover windows at night (blankets, bubble wrap, foil-faced foam board); open south windows on sunny days.
- Roll towels against door gaps.
- Shut off the propane water heater if not needed (heat water on the stove for washing instead).
- Keep the **tank lid, regulator, and path to the tank clear of snow** so it can be refilled and the regulator vent doesn't block.
- **Order propane early in the fall** — in a crisis, delivery may stop. Don't let it go below about 30% going into winter storms.

2. Reconnecting the wood stove to our chimney

We have a chimney and an old stove in the basement. This is a **great backup — but only if it is done right.** A bad connection can cause a **chimney fire, house fire, or CO poisoning**. Do it **now, while professionals are available**, so it's ready when we need it.

Step 1 — Inspect the stove

- Look for a **metal label** on the back (UL or other listing, brand, model, required clearances). If it has a label, the **label and owner's manual rule**.
- Check for **cracks** in the firebox (cast iron or steel), warped plates, missing firebricks, broken door glass, worn door gaskets (should seal tight — test by closing the door on a dollar bill; it should be hard to pull out).
- If cracked or badly worn: **replace it.** A new **EPA-certified stove** burns cleaner, uses less wood, and makes less creosote. *(EPA Burn Wise)*

Step 2 — Get the chimney inspected and swept

Hire a **CSIA-certified chimney sweep** (Chimney Safety Institute of America) for a **Level 2 inspection** (they often use a camera). They will check:

- **Cracks, missing mortar, or blockages** (bird nests are common).
- **Creosote** buildup — the black, tarry fuel that causes chimney fires.
- **The flue liner.** Many older masonry chimneys have **cracked clay tiles or no liner at all**. Stoves often need a **stainless-steel liner sized to the stove's flue outlet** (for example a 6-inch stove → a 6-inch liner). An oversized flue drafts poorly and builds creosote fast.
- **Is this flue shared** with the furnace or water heater? **A wood stove must have its own flue** unless the manufacturer and code specifically allow otherwise. Many propane appliances vent through the chimney — the sweep must check this.
- **Chimney height:** the top should be at least **3 ft above where it exits the roof** and **2 ft higher than anything within 10 ft** (the "3-2-10 rule"). Too short = poor draft and smoke.
- **Cap and spark arrestor** — important on the windy, dry plains.

Step 3 — The connection (stovepipe/connector)

- The pipe from the stove to the chimney is called the **connector**. It enters the chimney through a **thimble** (a clay or metal sleeve set in the masonry).
- **Single-wall stovepipe** needs **18 inches** clearance from combustibles (wood, drywall, joists) unless the stove or pipe label says otherwise. **Double-wall (close-clearance) pipe** allows less — follow its label. *(NFPA 211)*
- Keep the connector **short**, with few elbows. Horizontal runs should slope **up toward the chimney** (commonly about ¼ inch per foot).
- Crimped (male) ends point **down toward the stove** so creosote drips back inside.
- Fasten each joint with **3 sheet-metal screws**.
- **Never** run stovepipe through a wall, floor, or ceiling by itself. It must go through a listed thimble or wall pass-through.
- **Never connect into a fireplace opening without a proper liner and connection** — the old fireplace flue is usually far too big and unsafe.

Step 4 — Clearances and floor

- **Labeled stove:** use the clearances on the label/manual.
- **Unlabeled (unlisted) old stove:** NFPA 211 requires large clearances — **36 inches** from the stove to combustible walls/materials unless protected with approved shielding. Get the sweep or installer to confirm.
- **Floor:** a **noncombustible hearth pad** (stone, tile on cement board, or a listed pad) sized per the manual. A concrete basement floor usually works — but check that nothing combustible is under or near it.
- Nothing stored within 3 ft: no firewood stacks, laundry, kids' toys, or propane bottles.

Step 5 — Permits, insurance, test fire

- Ask **Morgan County building department** if a permit/inspection is required.
- **Tell our home insurance** — some require a professional install.
- **Draft test:** with a window cracked, hold a lit match or incense near the open stove door — smoke should be pulled **in**. If it's not, warm the flue first by burning a twist of newspaper. If it still smokes into the room, **stop and call the sweep.**
- Do a few small test fires **with CO alarms running** and someone watching.

Using the stove safely

- Burn only **dry, seasoned firewood** (split and dried at least 6 months, under 20% moisture — a cheap moisture meter checks this). *(EPA Burn Wise)* Wet wood = smoke, little heat, lots of creosote.

- **Never burn:** trash, plastics, treated or painted wood, pallets with "MB" stamp, cardboard boxes in bulk, or gasoline/charcoal lighter fluid to start it.
- Keep a **fireplace screen/safety gate** around the stove — **toddlers and the 6-year-old must never be near it.** A stove can cause severe burns in a second.
- **Ashes:** put in a **metal can with a lid**, set outside on dirt/concrete, away from the house. Ashes stay hot for days.
- **Sweep the chimney** at least once a year; check monthly when burning heavily.
- **Chimney fire signs:** loud roaring, popping, dense smoke, hot chimney. **Close the stove air controls, get everyone out, call 911.**
- Keep an **ABC fire extinguisher** near the stove and in the kitchen.

Firewood on the plains

Trees are scarce around Wiggins, so **stock up now**. Firewood is sold by the **cord** (a stack 4 ft × 4 ft × 8 ft = 128 cubic feet). How much we need depends on the house, stove, and winter — heating a whole house all winter can take **several cords**. Heating **one room** as an emergency backup takes much less. **Track it:** note how much one cold week uses, then multiply.

- Store wood **off the ground, covered on top, open on the sides**, at least 30 ft from the house when possible (fire safety), with a few days' supply closer.
- Scrap lumber (untreated), downed cottonwood along the river (with landowner permission), and tree trimmings can supplement.

If there were no chimney

(Not our case — we have one.) Never improvise a vent out a window or wall. A stove with no chimney needs a **professionally installed Class A insulated chimney pipe**.

3. Backup propane heaters

- **Only** use heaters **labeled for indoor use**, with an **oxygen depletion sensor (ODS)** and tip-over shutoff. Follow the manual about ventilation (most require a cracked window).
- Use with a **CO alarm**, never while sleeping, never near curtains or kids.
- **Never use for heat indoors:** gas oven or stovetop, charcoal or propane grills, camp stoves, patio heaters, kerosene "torpedo" heaters, or a generator. *(CDC)*
- Store 1-lb and 20-lb propane cylinders **outside**, never in the house or basement.

4. No heat at all — the "one warm room" plan

1. **Choose one room** — ideally an **interior room** with few windows, near the wood stove if it is running (the **basement** can work well: the earth around it holds steady temperature, and it's also our tornado shelter).
2. **Move everyone in**, including Grandma and Grandpa. Shared body heat matters.

3. **Insulate it:** hang blankets over doors and windows, put rugs/cardboard on the floor, tape plastic sheeting over windows.
4. **Make a "tent" inside the room:** a table covered with blankets, or a small camping tent, holds heat around sleepers.
5. **Layers:** wool or synthetic, not cotton. Hats, dry socks, mittens. Change damp clothes right away.
6. **Eat and drink:** the body makes heat from food. Warm drinks help (no alcohol — it makes hypothermia worse).
7. **Babies:** see Chapter 4 for safe warmth. Never put a baby to sleep with loose blankets, pillows, or in an adult bed.
8. **A tent inside the room** (The Prepared, after the 2021 Texas blackout): set up a camping tent with its rainfly on, pile blankets over it, and sleep inside. A person gives off roughly **330 BTU of heat per hour** — a small enclosed space traps it.
9. **Shovel paths often** during a storm (to the propane tank, coop, hutches, woodpile) rather than all at once at the end.
10. **Hot water bottles** (warm, not hot — wrap in a towel; never against a baby's skin).

Summer heat

Summers here can be very hot and dry. Babies and older adults overheat fast.

- Keep the house closed and shaded in the day, open it at night.
- Use the basement as the cool room.
- Wet cloths, cool (not cold) baths for kids, plenty of fluids.
- **Never leave a child or pet in a car.**
- Heat stroke (hot, confused, not sweating) = **911**.

Do it now

- ☐ Battery **CO alarms** on every level and sleeping area, both houses. Smoke alarms tested.
- ☐ **CSIA-certified sweep: Level 2 chimney inspection** + liner recommendation.
- ☐ Inspect/repair or replace the stove; professional connection; ask county about permit; tell insurance.
- ☐ Hearth pad, stove gate, metal ash can, ABC extinguisher, moisture meter.
- ☐ Buy firewood (dry) and store it right.
- ☐ Electrician: generator inlet + transfer switch/interlock (furnace, well, fridge).
- ☐ Fill propane in early fall; record gauge.
- ☐ Wool blankets, sleeping bags rated for cold, plastic sheeting, duct tape.

CHAPTER 4

Babies & Kids

Our kids are the most vulnerable people in the family. Babies **get cold, dehydrated, and sick faster** than adults, and they can't tell us. This chapter is deliberately careful. **When in doubt, call the pediatrician or 911 while phones work.**

Our kids at a glance

Kid	Feeds on	Biggest risks in an emergency
Newborn (born late Sep 2026)	**Breast milk only**	Cold, dehydration, fever/infection, unsafe sleep
1-year-old	Formula → moving to whole milk	Running out of formula/milk, choking, dehydration
2-year-old	Regular milk + table food	Dehydration from diarrhea, choking, burns (stove)
6-year-old	Regular food	Fear/stress, wandering, burns, cold exposure outdoors

The newborn — breastfeeding

Breastfeeding is the safest emergency food in the world: always the right temperature, no water needed, no bottles to clean, and it protects against infection. **Protect Mom's ability to nurse.**

- Newborns typically nurse **8–12 times per 24 hours**. *(AAP)* Keep nursing on demand, even during stress — the more baby nurses, the more milk is made.
- **Stress does not usually dry up milk**, but it can slow let-down. Skin-to-skin, a warm quiet spot, and a calm helper help.
- **The newborn needs NO water, juice, tea, or food** — only breast milk. Water can be dangerous for young babies.
- **Is baby getting enough?** After the first week: about **6 or more wet diapers a day**, regular stools, swallowing sounds while feeding, steady weight gain, alert when awake.
- Get a **manual breast pump** (no power needed) and learn **hand expression** — in case Mom and baby are briefly apart, or Mom is engorged.
- Pumped milk storage *(CDC)*: room temperature up to **4 hours**; refrigerator up to **4 days**; freezer about **6 months best (up to 12)**.

Feeding Mom (very important)

- A breastfeeding mother needs about **450–500 extra calories a day**. *(CDC)* **She eats first** when food is short — her milk feeds the baby.

- **Drink to thirst** — a simple habit is a full glass of water every time she nurses. Dark urine = drink more.
- Keep taking a **prenatal vitamin**; stockpile several months.
- Eggs and rabbit are great protein for her — add fat and carbs (see Chapter 6 on "rabbit starvation").
- If Mom gets sick (cold, flu, stomach bug), she can usually **keep breastfeeding** — her antibodies pass to the baby. Check any medicine with the doctor/pharmacist or LactMed (a National Library of Medicine database).

Backup if Mom can't nurse

If Mom were injured, very ill, or separated, the newborn would need **infant formula**. Keep a small backup supply of **ready-to-feed infant formula** (no water needed) and disposable nipples — but don't use it routinely, because skipping feeds lowers Mom's milk supply. Ask the pediatrician which formula.

The 1-year-old — formula and moving to milk

- The American Academy of Pediatrics says most children can switch from formula to **whole cow's milk at about 12 months**, about **16–24 oz (2–3 cups) a day**, plus solid foods. **Toddler/"transition" formulas are not needed.** *(AAP / HealthyChildren.org)* Talk to the pediatrician about switching **now**, before an emergency — it makes us less dependent on formula.
- Too much milk (over about 24 oz a day) can fill them up and lower iron. Offer iron-rich foods: **eggs, meat (including rabbit), beans, iron-fortified cereal.**

If still on formula — stockpile math

1. Check the can: it says **"makes about ___ fl oz."**
2. Daily ounces × days ÷ ounces per can = cans needed. *Example: 24 oz/day × 30 days = 720 oz. If a can makes 90 oz → 8 cans for a month.*
3. Rotate: use the oldest first, watch expiration dates.

Formula safety — never break these rules

- **NEVER make homemade formula.** It can cause serious illness and death. *(FDA, AAP)*
- **NEVER water down formula** to stretch it. It can cause dangerous low sodium (seizures) and malnutrition. *(AAP)*
- **NEVER give an infant under 12 months cow's milk, goat milk, or plant milk** as a formula substitute.
- Mix **exactly** as the label says.
- Use **safe water** (bottled, or boiled then cooled). **Ready-to-feed** is best when water is uncertain.

- Throw out prepared formula after the time on the label (often 1 hour once baby starts drinking).

The 2-year-old — milk and food

- Keeps drinking milk + eating family food. Ask the doctor whether to stay on whole milk or switch to lower-fat after age 2.
- **Shelf-stable milk:** stock **UHT (boxed) milk** and **powdered whole milk** for when fresh milk runs out. Once opened, refrigerate/use quickly.

Safe foods from our farm for the little ones

- **Eggs:** great food for all the kids except the newborn. **Cook fully** — firm yolks and whites — for young children. *(Eggs can be introduced to babies around 6 months; the newborn is far from that.)*
- **Rabbit meat:** cook thoroughly; shred or mince finely; **remove every small bone** — rabbit bones are thin and sharp.
- **Choking hazards for under 4:** whole grapes, nuts, popcorn, hot dog rounds, hard raw vegetables, chunks of meat, spoonfuls of peanut butter. Cut food into small, soft pieces.
- **No honey** for babies under 1 year (botulism).
- **Wash hands after handling chickens or eggs** — chickens commonly carry Salmonella. Kids under 5 should not kiss or snuggle chickens. *(CDC)*

Keeping babies warm (and safe)

- Dress the baby in **about one more layer than an adult** would wear to be comfortable. *(AAP)*
- **Feel the chest or back of the neck** — warm and dry is right. Cold hands/feet alone are normal; sweaty or flushed = too hot.
- **Skin-to-skin** with an awake adult, covered by a blanket, is an excellent way to warm a baby.
- **Safe sleep, every sleep** *(AAP)*: on the **back**, on a **firm, flat surface** (crib, bassinet, or play yard) **in the same room** as parents, with **no loose blankets, pillows, or bumpers**. Use a **wearable blanket/sleep sack** and footed pajamas for warmth instead of loose blankets. Don't let baby sleep in a car seat outside the car, a swing, or an adult bed or couch.
- Keep babies **away from the wood stove and space heaters** — and keep the sleep space away from them too.
- **A baby who is cold to the touch, unusually quiet, floppy, or not feeding = emergency.**

Diapers and hygiene

- Rough daily use: **newborns about 8–12**, older babies/toddlers fewer. Stock **at least a month** of each size the kids will be in (sizes change fast!).

- **Backup: cloth diapers** (prefolds + covers) and a bucket with lid. Washing needs water and heat — plan for it.
- Wipes, diaper cream, garbage bags, and hand sanitizer.
- Bury or bag soiled diapers away from water and gardens; never in chicken compost.

Dehydration — know the signs

Kids can get dangerously dry from **diarrhea, vomiting, fever, or heat**.

Warning signs:

- Fewer wet diapers (or none in 6+ hours for a baby), dark urine
- Dry mouth, **no tears** when crying
- **Sunken soft spot** on top of a baby's head, sunken eyes
- Very sleepy, floppy, or very fussy

What to do: keep breastfeeding/feeding; give **oral rehydration solution** (store-bought such as Pedialyte) in small sips for kids over infancy as directed on the package and by the doctor. **A newborn with any of these signs needs a doctor right away.**

Fever

- **Any baby under 3 months with a temperature of 100.4°F (38°C) or higher is an emergency — call the doctor or go to the ER immediately.** *(AAP)* That is our newborn right now.
- Get a **digital thermometer** (rectal is most accurate for babies).
- Ask the pediatrician **now** for the correct dose of acetaminophen (and ibuprofen for 6+ months) by weight, and write it in the Home Worksheet. **Never give aspirin to children.**

The 6-year-old (and keeping kids steady)

- Kids handle hard times better with **routine, honest simple explanations, and jobs**. Give the 6-year-old real jobs: gathering eggs, feeding rabbits, carrying kindling (supervised), being "radio helper."
- Teach: our address, parents' names and phone numbers, the meeting places, "stay away from the stove and the road," and to **never go outside in a blizzard alone**.
- Keep books, crayons, cards, and favorite toys in the emergency kit.
- Watch for stress: bedwetting, nightmares, clinginess. It's normal — extra hugs and routine help.

Do it now

- [ ] Ask the pediatrician: switching the 1-year-old to whole milk; medicine doses by weight; what to keep in a baby kit.
- [ ] Manual breast pump + bottles + milk storage bags.
- [ ] Small backup of **ready-to-feed infant formula** (newborn) + 1 month of the 1-year-old's formula/milk plan.
- [ ] UHT boxed milk and powdered whole milk.
- [ ] 1 month of diapers in the next sizes + cloth backup, wipes, cream.
- [ ] Prenatal vitamins (3+ months) for Mom.
- [ ] Digital thermometer, infant/children's acetaminophen, oral rehydration solution.
- [ ] Wearable blankets/sleep sacks, footed pajamas, a portable play yard for the warm room.
- [ ] Stove gate.

CHAPTER 5

Food: Eggs & Chickens

Our hens give **6–12 eggs a day** — that's real food: each large egg has about **70 calories and 6 grams of high-quality protein** *(USDA)*. Nine eggs a day ≈ **630 calories and 55 g protein** for the family. The challenge is **keeping the hens fed and laying when we can't buy feed, especially in winter.**

How much hens eat and drink

- A laying hen eats roughly **¼ lb of feed a day** (more in cold weather). *(University Extension, typical figure)*
- 15 hens ≈ 3¾ lb/day ≈ **110 lb per month**.
- A hen drinks roughly **a pint of water a day**, more in heat. **No water = no eggs within a day or two.**
- Laying hens need feed of about **16% protein** plus **calcium** for strong shells. *(Extension)*

Stockpile now: several months of **layer feed** in metal cans with lids (mice!), plus **oyster shell**. Feed goes stale — rotate, use oldest first, keep dry. **Never feed moldy feed.**

Why hens stop laying in winter

1. **Short days.** Hens lay best with about **14–16 hours of light**. In December, Wiggins gets only about 9½ hours of daylight. *(Extension)*
2. **Molting** in fall — they stop to regrow feathers (needs extra protein).
3. **Cold stress, not enough feed or water.**
4. **Age** — production drops after the second year.

What helps:

- **Add light** to make about 14 hours total: a low-watt LED on a timer, turned on **early morning** (so they go to roost at natural dusk). If power is out, a small solar light kit can do this.
- **Raise protein** during molt (insects, cooked egg, meat scraps — see below).
- **Keep water unfrozen.** Heated waterer base if power; otherwise swap waterers 2–3 times a day with warm water.
- **Plan to replace hens:** hatch or raise new pullets each spring so we always have young layers. A rooster lets us hatch our own (a broody hen or incubator).

Winter coop care on the plains

- **Ventilation without drafts:** vents high up, no wind blowing on roosting birds. Damp air causes **frostbite on combs** more than cold does.

- **Deep litter:** keep adding dry straw/pine shavings; it composts slowly and adds a little warmth.
- **Avoid heat lamps** — they are a leading cause of coop and barn fires. Chickens handle cold well if dry and out of the wind.
- **Wind block:** straw bales, tarps, or plywood on the north and west sides.
- Close them in **securely at night**: coyotes, foxes, raccoons, skunks, and owls are all around Wiggins. Hardware cloth, not chicken wire.

The "Raken" house — rabbits over chickens (Joel Salatin)

At Polyface Farm, Joel Salatin's family hangs **rabbit cages above a floor where hens run** — the "Raken" (rabbit + chicken) house. Rabbit droppings and urine fall into **deep carbon bedding** (wood shavings, sawdust, leaves, chopped straw). The hens scratch it constantly, which keeps it from smelling, turns it into compost, and **breeds worms and bugs that the hens eat — free protein all winter.**

Salatin's deep-bedding rules:

- Start with **at least 8 inches, preferably 12+**, of carbon bedding.
- **Add more carbon whenever it smells.** Long hay/straw mats — chop it or add a little at a time.
- A dirt floor drains best. Good ventilation, especially in warm weather.
- Toss scratch grain on the bedding to get the hens working it.
- The finished compost is excellent garden fertilizer in spring.

This fits our place perfectly: it **puts Daniel's two animals together** so the rabbits help feed the chickens without killing a single rabbit, and it keeps both in one sheltered, easier-to-water building in winter. *(Keep rabbit cages high and solid enough that hens can't roost on them and soil the rabbits.)*

Feeding chickens when feed runs out

Chickens are **omnivores** — they eat grains, greens, seeds, bugs, and meat. Mix several of these:

Grains (energy)

- **Whole or cracked wheat, corn, milo (sorghum), barley, oats.** We live in grain country — buy or trade directly from local farmers and elevators **now**, by the bag or ton. Grain stores for years if dry and sealed.
- Corn alone is **low in protein** — hens will lay poorly on corn only. Add protein.

Protein (eggs need protein)

- **Insects** (see the bug farms below).
- **Cooked eggs** (scrambled, with shell crushed in) — especially cracked or extra eggs.

- **Cooked rabbit scraps and organs** (see below).
- Fish scraps, cooked; milk/whey or cottage cheese in small amounts.
- **Dried peas** or other legumes — **cooked**. Raw dry beans are toxic.

Calcium (shells)

- **Oyster shell** free-choice; or **our own eggshells** — rinse, **bake dry**, crush fine, offer in a separate dish.

Greens (vitamins, especially in winter)

- Stored **pumpkins, squash, cabbage, beets, carrots** (grow extras in summer for the chickens).
- **Alfalfa hay leaves** (shake the leafy bits out of a bale).
- **Sprouted grain "fodder"**: soak wheat or barley overnight, spread in trays with drain holes, rinse twice a day, feed in **about 6–8 days** when it's a green mat. *Honest note:* sprouting does **not** add calories (it uses some up), but it gives **fresh greens and vitamins in winter** and hens love it. Needs room temperature and water.

Kitchen scraps

- **OK:** vegetable and fruit scraps, cooked rice/pasta/oats, bread (small amounts), cooked meat scraps, garden waste.
- **Not OK: moldy or rotten food**, **avocado** (pits/skins), **raw or dried beans**, **chocolate**, very **salty** food, **green potato peels**, onions in large amounts, coffee grounds.

Bug farms — protein we can raise indoors in winter

Mealworms (easiest for winter, indoors)

- Plastic bins (with ventilation holes) of **wheat bran or rolled oats** a few inches deep.
- Add **slices of carrot or potato** for moisture (replace before mold).
- Keep at **room temperature** — the life cycle (beetle → egg → worm → pupa → beetle) takes a few months, slower when cool.
- Start in **late summer** so the colony is big by winter. Harvest larger worms; leave enough to breed.
- Start with a few thousand bought live mealworms.

Black soldier fly larvae (BSFL) — best in summer

- Larvae eat almost **any food waste and manure** and are high in protein and fat. The mature larvae crawl out on their own into a bucket ("self-harvesting" bins).

- **They need warmth** (they slow down a lot in cool weather) — great from late spring to early fall outdoors. Over winter, they're hard to keep without a heated space. Plan for summer production and **dry the extra larvae** (in an oven or dehydrator) to store for winter.

Earthworms (red wigglers)

- A worm bin with shredded cardboard, kitchen scraps, and rabbit manure (rabbit manure is great for worms). Keep it above freezing — basement works.

Daniel's idea: rabbits as chicken food

Yes — this works, with the right approach. Chickens love meat and it's real protein, especially for winter laying and molting.

Do it this way:

- **We eat the rabbit meat.** Feed the chickens the parts we don't eat: **organs we're not using, heads, scraps, fat trimmings, and bones (cooked, then cracked)**.
- **Cook it first.** Boil scraps and organs until fully cooked. Cooking kills parasites and germs (including **tularemia**, which rabbits can carry) and keeps the hens from developing a taste for raw meat that can lead to egg- or flock-pecking.
- **Never feed spoiled or sick-animal meat.** Bury or burn any rabbit that died of illness.
- Feed scraps the same day, a modest amount, alongside grain — not as the only food.

The energy trade-off (important): every time food passes through an animal, **most of its energy is lost**. A rabbit turns hay into meat inefficiently; a hen turns rabbit meat into eggs inefficiently. **Feeding whole rabbits to chickens to get eggs yields far fewer calories than eating the rabbit ourselves.** So the best plan is: **people eat the meat; chickens get the leftovers.** That's a win twice.

About the reverse: **rabbits are strict plant-eaters** — never feed chicken or any meat to rabbits.

Keeping eggs

- **Unwashed** eggs straight from clean nests keep their natural coating. The USDA recommends refrigeration for safety; in an outage keep eggs in the coolest place possible and use oldest first.
- **Freeze extra eggs:** crack, beat lightly, freeze in portions (ice cube trays). Don't freeze in the shell.
- Traditional preservation (like "water-glassing" in pickling lime) is popular but **not tested/approved by the USDA** — use at your own risk; discard any egg that smells off or floats suspiciously.
- **Cook eggs fully** for the kids, Grandma and Grandpa, and Mom.

Flock health and bird flu

- **Avian influenza (bird flu)** has hit Colorado poultry and dairies, including in nearby Weld County. Keep wild birds out of feed and water, don't touch sick or dead wild birds, wash hands, use dedicated boots for the coop.
- **Sudden deaths, swollen heads, purple combs, or a big drop in eating/laying:** don't eat those birds or eggs; report to the **Colorado Department of Agriculture State Veterinarian's office**.
- If you butcher an old hen for soup: only healthy birds, cook thoroughly.

Do it now

- [ ] Stockpile 3+ months of layer feed (sealed metal cans) + oyster shell.
- [ ] Line up a local farmer/elevator for bulk wheat, corn, or milo.
- [ ] Coop light on a timer; heated waterer (+ manual backup).
- [ ] Start a mealworm bin now; plan BSFL bin for next summer.
- [ ] Grow extra squash/pumpkins/beets next summer for winter greens.
- [ ] Get a rooster or plan spring chicks to keep young layers coming.
- [ ] Predator-proof the coop with hardware cloth.

CHAPTER 6

Rabbits & Feeding Eight People

Rabbits are one of the best survival meat animals: quiet, small, fast-breeding, and they turn hay and greens into meat. **But you cannot live on rabbit alone** — the meat is very lean. This chapter covers raising, butchering, storing, and *how rabbit fits into a full diet for 8 people*.

Rabbit basics

Fact	Typical figure *(Extension guides, meat breeds)*
Pregnancy (gestation)	about **31 days** (28–35)
Litter size	often **6–8 kits** (varies a lot)
Weaning	about **4–8 weeks**
Butcher age ("fryers")	about **8–12 weeks**, around 4–5 lb live weight
Dressed meat	a bit over **half** the live weight (with bone)
First breeding (medium breeds)	does around **5–6 months** old

A breeding plan

- Keep a few good **does** (females) and **1–2 bucks** (males), housed separately.
- **Always take the doe to the buck's cage** (does defend their space).
- Put a **nest box** with straw/hay in at **day 27–28**. The doe pulls fur to line it.
- A moderate, sustainable plan: rebreed the doe when kits are about 4–6 weeks old, giving about **4–5 litters a year** per doe. Rest does that look thin.
- Keep records: doe, buck, breeding date, kindling date, number of kits, weaned, butchered.
- **Avoid inbreeding:** bring in an unrelated buck every year or two (trade with neighbors).

Rough output: 4 does × 4–5 litters × ~6 kits raised ≈ **about 100 fryers a year** — roughly **one rabbit every 3–4 days**. Great protein, but not enough calories alone for 8 people (see math below).

Feeding rabbits — especially in winter

- **Unlimited grass hay** (or grass/alfalfa mix) is the base of the diet. **Stock hay before winter** — it's cheap here in fall and scarce after a bad winter. Track how fast we use it.
- **Pellets** if available (stockpile in sealed cans). If pellets run out: hay + **oats/barley** (small amounts) + greens.

- **Safe greens and extras:** dandelions, plantain, clover, grass, carrot tops, kale, cabbage (small amounts), sunflower heads, sprouted grain fodder, tree branches (apple, willow — rabbits love to chew).
- **Avoid:** moldy hay, lawn clippings that have fermented, potato leaves, rhubarb, nightshade, avocado, onions, ice-cold or spoiled food, and **any meat**.
- **Water is the #1 winter problem.** Bottles freeze and crack — use **heavy crocks** and swap with warm water 2–3 times a day. Nursing does drink a lot.

Winter and summer care

- Rabbits handle **cold** well if **dry and out of the wind**. Wrap hutches on the north/west, add a roof overhang, and give extra hay to burrow into.
- **Winter kits are at risk** of freezing. Give a deep, well-bedded nest box; move the nest box (or the doe's cage) to a sheltered spot like a garage or barn in a cold snap. Or plan breeding so most kits are born in milder months.
- **Heat is the bigger killer** — rabbits suffer above about 85°F. In summer: shade, airflow, frozen water bottles to lean against.

Housing tip

Consider the **Raken house** (Chapter 5): rabbit cages hung above the chicken floor. The rabbits' manure feeds the deep bedding the hens scratch for bugs, and one building is easier to keep out of the wind and keep water thawed.

Butchering (humane and clean)

1. **Withhold feed** about 12 hours before (keep water).
2. Dispatch quickly and humanely (learn from an experienced person or a CSU Extension class — **practice now**).
3. Hang, skin, and gut on a clean surface. Keep the bladder and intestines intact.
4. **Check the liver:** it should be smooth and dark red. **White spots** may mean parasites/disease — throw out the liver; if the whole carcass looks unhealthy, discard it.
5. **Wear gloves.** Rabbits can carry **tularemia**, a bacterial disease. Domestic rabbits are low risk; **wild rabbits** acting sick or sluggish — don't handle or eat. *(CDC)*
6. Chill the meat quickly (cold water bath or fridge).
7. **Organs and scraps → cook for the chickens** (Chapter 5). Hides can be tanned; manure goes to the garden (rabbit manure can go straight on beds).

Storing rabbit meat

- **Fridge:** cook within 1–2 days.

- **Freezer:** months. Wrap tightly to prevent freezer burn.
- **Winter:** a cold snap is a natural freezer — but meat outside must be protected from animals and kept **solidly frozen**, not thawing in sun.
- **Canning:** meat **must be pressure-canned** using a tested USDA/Extension recipe. **Never water-bath can meat** — botulism risk.
- **Jerky:** USDA recommends heating meat to **160°F** before or after drying for safety.
- **Cooking:** cook rabbit thoroughly — to **160°F** internal temperature. *(USDA)*

⚠ "Rabbit starvation" (protein poisoning)

Rabbit meat is **very lean** — mostly protein, little fat. People who eat **only lean meat** for days to weeks get sick: **nausea, diarrhea, weakness, and constant hunger**, and can actually starve while eating plenty. Your body can only safely handle a limited share of calories from protein; the rest must come from **fat and carbohydrates**.

Always pair rabbit with:

- **Fat:** cooking oil, lard, butter/ghee, bacon grease, peanut butter, nuts, **egg yolks**.
- **Carbs:** rice, wheat/flour, oats, potatoes, beans, corn, squash.
- Example meal: rabbit stew with potatoes, beans, and a big spoon of lard or oil; eggs on the side.

Calorie math for our 8 people

These are **rough daily estimates** based on the USDA Dietary Guidelines' calorie tables (moderately active), adjusted for breastfeeding:

Person	Approx. calories/day
Daniel (physical work)	2,800
Wife (breastfeeding: +450–500)	2,500
Grandpa	2,200
Grandma	1,800
6-year-old	1,400
2-year-old	1,000
1-year-old (part from milk)	900
Newborn	breast milk only
Family total	**≈ 12,600 per day**

Work in the cold burns more. These are estimates, not prescriptions.

What the farm provides

- **Eggs:** ~9 eggs/day × 70 cal ≈ **630 cal/day** (less in winter).
- **Rabbit:** one fryer every 3–4 days. A fryer gives roughly 1,000 calories of meat. *(USDA lists raw rabbit at about 136 cal per 100 g.)* ≈ **300 cal/day**.
- **Together: about 1,000 calories/day — under 10% of what we need.** Eggs and rabbits are **excellent protein**, but **stored staples must supply most of the calories.**

What we'd need from storage (per month)

12,600 cal × 30 days ≈ **378,000 calories**. Roughly:

Staple	Approx. calories per lb (dry)	One month for us
Rice / wheat / flour / oats	~1,600	**~150 lb**
Dry beans / lentils	~1,550	**~40 lb**
Cooking oil / lard	~4,000	**~2 gallons** (≈ 15 lb)
Eggs + rabbit (farm)	—	~30,000 cal

Plus: salt, sugar/honey, powdered milk, canned vegetables and fruit, vitamins, spices, and comfort foods for the kids. See Chapter 7.

Do it now

- [ ] Count does, bucks, ages; start breeding records.
- [ ] Stock hay for the full winter + sealed pellets.
- [ ] Heavy water crocks for winter.
- [ ] Learn butchering hands-on (CSU Extension / experienced neighbor).
- [ ] Pressure canner (and a tested meat canning guide) or freezer space.
- [ ] Store fats: oil, lard, peanut butter.

CHAPTER 7

Food Storage, Gardening & Foraging on the Plains

Chapter 6 showed that **eggs and rabbits supply protein, but stored staples must supply most of our calories.** This chapter covers what to store, how to cook without power, how to garden in our short, windy, hail-prone season, and what we can realistically find, hunt, or fish around Wiggins.

Storage principles

1. **Store what we already eat**, and eat what we store (rotate: new in back, old in front).
2. **Cool, dark, dry, and rodent-proof.** Basement shelves are good. Off the floor.
3. Bulk grains and beans: **food-grade 5-gallon buckets** with **Mylar bags and oxygen absorbers**, or sealed metal cans. Stored this way, dry white rice, wheat, and dry beans keep for many years (Extension guidance).
4. Write the **date** on everything.
5. **Manual can opener** (two!).

How much — the LDS (Latter-day Saints) method

The Church of Jesus Christ of Latter-day Saints has taught home food storage for generations, and its method is simple and well tested:

1. **Start with one week** of the food you normally eat (what you buy for a day × 7).
2. Grow it to a **three-month supply** of normal foods, rotated.
3. Then build a **longer-term supply of basics that keep you alive: wheat, white rice, and beans**. Properly packaged and kept cool (75°F or below) and dry, these can last **30 years or more**. Cooking oil needs rotating every 1–2 years.

Their one-year "basics" per adult (Church guidance): **400 lb grains, 60 lb legumes, 16 lb powdered milk, 10 qt cooking oil, 60 lb sugar or honey, 8 lb salt**, plus 2 weeks of water. Children get a share: **3 and under 50%, 4–6 70%, 7–10 90%**, and they suggest using **next year's age** for growing kids. **Nursing babies share Mom's portion.**

For our family (4 adults + 6-year-old at 90% + two little ones at 50% each ≈ **5.9 adult shares**):

Basic	Per month	One year
Grains (wheat, rice, oats, corn, pasta)	~195 lb	~2,360 lb
Legumes (beans, lentils, split peas)	~30 lb	~355 lb
Powdered milk	~8 lb	~95 lb
Cooking oil	~5 qt	~59 qt (~15 gal)
Sugar or honey	~30 lb	~355 lb
Salt	~4 lb	~47 lb

These line up with our calorie math in Chapter 6. A year is a big goal — three months is a great start.

Warning (LDS/BYU guidance): only store **dry foods (about 10% moisture or less)** in airtight containers with oxygen absorbers — moist foods sealed without oxygen can cause **botulism**. Nuts, whole-wheat flour, brown rice, and jerky don't keep long-term.

A balanced pantry (starter targets for one month — multiply for more)

Group	Examples	Month target (approx.)
Grains	wheat berries + hand grain mill, white rice, oats, flour, pasta, cornmeal	150 lb
Beans/legumes	pinto, black beans, lentils, split peas	40 lb
Fats	vegetable oil, lard, shortening, peanut butter, nuts	2 gal oil + extras
Milk	UHT boxed milk, powdered whole milk	see Chapter 4
Canned	vegetables, fruit, tomatoes, soups, tuna, chicken	a can or two per person per day
Sugar/sweet	sugar, honey (not for newborn), jam	10–20 lb
Salt & seasoning	salt (also for preserving), spices, bouillon	10+ lb salt
Kids	baby cereal (iron-fortified), pouches, crackers, favorite snacks	as needed
Vitamins	multivitamins, prenatal vitamin	3+ months

Cooking without power

- **Wood stove top** (once reconnected) — great for pots of beans, stew, and heating water.
- **Propane camp stove or grill — OUTDOORS ONLY.** Never indoors or in a garage: carbon monoxide.
- **Thermal cooking:** bring beans/rice to a boil, then wrap the pot in blankets or put it in a cooler — it keeps cooking for hours with no fuel.
- Soak beans overnight to save fuel.

Fridge and freezer in an outage *(USDA)*

- Keep doors **closed**. Full freezer: ~**48 hours**; half full: ~**24 hours**. Fridge: ~**4 hours**.
- In winter, move food to coolers outside in the shade (protected from animals) when it is below freezing.
- **"When in doubt, throw it out"** — especially meat, dairy, eggs, leftovers above 40°F for more than 2 hours.

Gardening on the eastern plains

Our challenges: **short season, late spring and early fall frosts, hail, wind, alkaline soil, and limited water.**

- **Frost dates:** around Wiggins the average last spring frost is roughly **mid-May** and the first fall frost roughly **late September to early October**. Check **CSU Extension / Colorado Climate Center** for exact local dates.
- **Best calorie crops for us:** potatoes, winter squash and pumpkins (store all winter — and feed chickens), dry beans, corn, sunflowers (seeds for people, chickens, and rabbits).
- **Fast greens:** lettuce, spinach, kale, radishes, peas (cool season — plant early spring and again late summer).
- **Protect from hail and wind:** hail cloth or row covers on hoops, windbreak fences, raised beds.
- **Season extenders:** cold frames, low tunnels, a small hoop house — can stretch greens well into fall.
- **Water:** drip irrigation and thick mulch (straw). **Colorado law** lets homeowners collect rooftop rainwater in **up to two rain barrels totaling 110 gallons**, for outdoor use. Domestic well permits may limit outdoor watering — check ours.
- **Rabbit and chicken manure** = free fertilizer (compost chicken manure first; rabbit manure can go on directly).
- **Save seeds** from open-pollinated (heirloom) varieties. Keep a stash of seeds in a cool dry place — buy extra now.
- **Root cellar / basement storage:** potatoes, squash, carrots, onions, and cabbage keep for months in a cool, dark, slightly humid space.

Foraging around Wiggins — realistic expectations

The plains do **not** offer much wild food, especially in winter. Foraging is a **supplement**, not a plan. **Never eat a plant unless you are 100% sure what it is.** Don't feed foraged plants to the kids or Mom unless you are certain.

Common, useful wild foods here (learn them with a guide or CSU Extension class):

Plant	Where	Season	Notes
Dandelion	yards, fields	spring–fall	leaves, flowers, roots all edible
Lamb's quarters	gardens, disturbed ground	summer	like spinach; cook
Purslane	gardens, sidewalks	summer	eat raw or cooked
Curly dock	ditches, fields	spring	cook; contains oxalates — moderate amounts
Wild plum, chokecherry	river bottoms, fencerows	late summer	**chokecherry pits, leaves, and stems are poisonous** — eat only the pitted, cooked fruit (jelly, syrup)
Prickly pear cactus	dry grassland	summer–fall	pads and fruit; burn/scrape off the tiny spines (glochids)
Cattail	ponds, ditches, river	spring–summer	shoots and roots; only from **clean** water
Wild asparagus	irrigation ditches	spring	
Jerusalem artichoke (sunflower tubers)	ditches, field edges	fall–winter	tubers dug after frost

⚠ Deadly look-alikes near the South Platte

- **Water hemlock** — grows in wet places along rivers and ditches; **one of the most poisonous plants in North America.** Looks like wild carrot/parsnip. **Don't eat any wild plant in the carrot family.**
- **Poison hemlock** — common in Colorado ditches; purple-spotted stems; deadly.
- **Death camas** — onion-like bulbs with no onion smell.
- **Wild mushrooms** — don't, unless an expert identifies them.

Hunting and fishing

The area has **mule deer, white-tailed deer, pronghorn, pheasant, quail, doves, waterfowl, cottontails, and fish** (South Platte, Jackson Lake, area reservoirs).

- **Colorado Parks and Wildlife (CPW) licenses and seasons apply** — buy hunting and fishing licenses and learn the seasons **now**. Laws still apply in emergencies unless officials say otherwise. Hunt only where you have **landowner permission** or on public/walk-in access land.
- **Chronic Wasting Disease (CWD)** is present in Colorado deer and elk. CPW recommends **testing** animals and not eating meat from animals that test positive or look sick.
- Cook wild game thoroughly (**160°F** for ground meat).

- **Firearm safety with four kids:** guns **locked and unloaded**, ammunition stored separately, **keys/combination only known to adults.** Teach the 6-year-old: "Don't touch. Leave the room. Tell an adult."

Do it now

- [ ] Build the one-month pantry, then two, then three. Buckets, Mylar, oxygen absorbers.
- [ ] Hand grain mill + wheat berries; two manual can openers.
- [ ] Buy extra seeds; plan next year's garden (potatoes, squash, beans, corn).
- [ ] Two rain barrels (110 gal total).
- [ ] Hail cloth/row covers and a cold frame.
- [ ] CPW hunting & fishing licenses; learn a few local wild plants with an expert.
- [ ] Gun safe / lock boxes.

CHAPTER 8

Medical & First Aid

Get training now: a **Red Cross or American Heart Association class in First Aid and CPR — including infant and child CPR and choking.** It is the single best thing to do for this chapter. This book is a reminder, not a replacement.

> **Home remedies, fever medicine doses by weight, and local dangers (snakes, hantavirus, tularemia) are in Chapter 9.**
>
> **Call 911 / go to the ER while you can** for: trouble breathing, chest pain, a baby under 3 months with fever 100.4°F+, a child who is limp/unresponsive or won't wake normally, seizures, severe bleeding, serious burns, suspected CO poisoning, or signs of stroke (face drooping, arm weakness, speech trouble).

Hypothermia (body too cold) *(CDC)*

Signs — adults: shivering, exhaustion, confusion, fumbling hands, slurred speech, drowsiness. **Infants:** bright red, cold skin and very low energy.

What to do:

1. Get them into a **warm room or shelter**. Remove **wet** clothes.
2. **Warm the center of the body first** — chest, neck, head, groin — with dry blankets, layers, or **skin-to-skin** contact under loose blankets.
3. **Warm drinks** if fully awake (no alcohol). Never give drinks to someone who is unconscious.
4. Keep them dry and wrapped, including head and neck.
5. **Get medical help.** If there is no breathing or pulse, start **CPR**.

Older adults and babies get hypothermia indoors in cold houses — check Grandma, Grandpa, and the kids often.

Frostbite *(CDC)*

Signs: skin that is red then **white or grayish-yellow**, **firm or waxy**, and **numb**. Fingers, toes, ears, nose, cheeks.

What to do:

- Get to warmth. **Check for hypothermia first** (it's more urgent).
- **Don't walk on frostbitten feet** if you can avoid it.
- Rewarm in **warm — not hot — water** (comfortable to an unaffected person's hand), or with body heat (fingers in armpits).

- **Don't rub or massage**, and **don't use a stove, heating pad, or fire** to warm — numb skin burns easily.
- **Don't rewarm if it could freeze again** before reaching shelter — refreezing does more damage.
- Get medical care.

Carbon monoxide poisoning *(CDC)*

Signs: headache, dizziness, weakness, upset stomach, vomiting, chest pain, confusion. **Several people (or pets) feeling sick at once indoors** = think CO. Sleeping people can die without waking.

Action: Everyone out to fresh air now. Call 911. Don't go back in until it's cleared. Find and fix the source (generator, heater, stove, blocked chimney).

Burns

- **Cool running water for about 20 minutes** (not ice, not butter). Remove rings/tight items.
- Cover loosely with clean plastic wrap or a clean, non-fluffy cloth.
- **ER for:** burns on a baby or young child (any size beyond very small), the face, hands, feet, genitals, or joints, or burns larger than the person's palm, or blistered/white/charred skin.

Bleeding

- **Firm, direct pressure** with cloth for at least 10 minutes without peeking.
- Severe bleeding from an arm or leg that won't stop: a **tourniquet** (learn how in a "Stop the Bleed" class) 2–3 inches above the wound, tightened until bleeding stops; note the time.
- Clean small wounds with clean water, cover, watch for infection (spreading redness, warmth, pus, fever).

Choking

- **Infants (under 1):** 5 back blows, then 5 chest thrusts with two fingers — repeat. **Learn in a class.**
- **Children and adults:** abdominal thrusts (Heimlich).
- Call 911 if not cleared; CPR if they become unresponsive.

Vomiting and diarrhea

- **Hydration is everything.** Oral rehydration solution in small, frequent sips. Keep breastfeeding.
- Strict **handwashing**, separate towels, bleach-clean surfaces.

- Get help for: signs of dehydration (Chapter 4), blood in stool, high fever, or a baby who won't feed.

Fever in children

- **Under 3 months, 100.4°F+ → emergency, always.**
- Older kids: watch behavior more than the number. Fluids, rest, fever medicine **by weight per the doctor's chart**. Get help if they're very hard to wake, have trouble breathing, a stiff neck, a rash that doesn't fade when pressed, or fever lasting more than a couple of days.

Medicines and the family kit

- **Ask doctors now for extra supplies** of everyone's prescriptions (especially Grandma and Grandpa's). Keep a written list of meds and doses.
- Store medicines cool, dry, **locked away from kids**.

Family first aid kit:

- Bandages of all sizes, gauze, rolled gauze, medical tape, elastic wrap
- Tourniquet (commercial), trauma shears, tweezers, gloves
- Digital thermometer, saline, burn gel, antibiotic ointment
- Pain/fever medicine — adult and children's/infant (doses written on the box)
- Oral rehydration solution, antihistamine (adult and children's)
- Reading glasses, extra contacts/glasses, hearing aid batteries
- Instant cold packs, emergency (Mylar) blankets
- This book and the Red Cross First Aid app (downloaded while we have signal)

Do it now

- [ ] First Aid + infant/child CPR class (Daniel, wife, Grandma, Grandpa).
- [ ] Stop the Bleed class.
- [ ] Extra prescriptions; written med list in the Home Worksheet.
- [ ] Build the first aid kit; check expiration dates each fall.

CHAPTER 9

Home Remedies: Wounds & Fevers

This chapter is for when **someone is hurt or sick and the doctor is far away, slow to reach, or not available**. It covers what really works at home for cuts, burns, and fevers, which "natural" remedies are worth trying (and which aren't), and the dangers specific to life on a farm near Wiggins.

> **Honest rule of this chapter:** the remedies that work best are plain ones — **clean water, pressure, honey, fluids, rest, and the right dose of the right medicine.** Herbs are listed with how strong the evidence really is. **No home remedy replaces the ER for the "Get help NOW" signs below.**

Our nearest care

- **Emergency: call 911.** In bad weather, ambulances may be slow — call early.
- **Nearest ER: CommonSpirit St. Elizabeth Hospital**, 1000 Lincoln St, **Fort Morgan** (formerly Colorado Plains Medical Center). 24-hour ER, Level III trauma center. Main phone **970-867-3391**. The ER entrance is on **Grant Street, east side** of the hospital. *(Hospital website, checked Oct 2026.)* Serious injuries, snakebites, or very sick babies may be transferred on to Greeley or Denver.
- **Poison Control: 1-800-222-1222** — free, 24/7, for poisonings, plants, bites, medicine mistakes. Also answers "how much is too much" medicine questions.
- **Nurse advice lines:** ask the pediatrician's office and your insurance for their 24-hour nurse line, and write it in the Home Worksheet.

⚠ Get help NOW — don't try to treat these at home

Anyone:

- Bleeding that **won't stop after 10 minutes of hard pressure**, or spurting blood
- Wounds that are **deep, gaping, over a joint, on the face, or have something stuck in them**
- **Animal bites to the face or hands**, deep punctures, or any bite from a **wild animal** (rabies risk — skunks, bats, raccoons, foxes are the main carriers in Colorado)
- **Rattlesnake bite** (see below)
- Burns that are **larger than the palm**, on the face/hands/feet/genitals, or white, leathery, or charred
- **Red streaks** spreading from a wound, fever with a wound, or a wound that smells foul
- Trouble breathing, chest pain, confusion, a seizure, or fainting
- Signs of stroke: face drooping, arm weakness, speech trouble

Babies and kids:

- **Any fever of 100.4°F (38°C) or higher in a baby under 3 months** (our newborn) — **go to care now. Don't give medicine first unless a doctor tells you.** *(AAP)*
- A child who is **limp, hard to wake, or can't be comforted**
- **Signs of dehydration** (no wet diaper in 6+ hours for a baby, no tears, sunken eyes or soft spot)
- Fever with **stiff neck, a purple rash that doesn't fade when pressed**, or trouble breathing
- **Fever over 104°F**, or any fever lasting more than **2–3 days**
- Any fever in a child with **a seizure**

Wounds — cuts, scrapes, and punctures

1. Stop the bleeding

1. **Press hard** directly on the wound with a clean cloth or gauze. **Hold for 10 minutes without peeking.** If blood soaks through, add more cloth on top — don't remove the first layer.
2. Raise the injured part if it's easy.
3. **Life-threatening bleeding** from an arm or leg that won't stop with pressure: use a **commercial tourniquet** 2–3 inches above the wound, tighten until bleeding stops, note the time, and get to the ER. *(Stop the Bleed / American Red Cross)*

2. Clean it well — the most important step

Infection, not the cut itself, is what makes small wounds dangerous when help is far away.

- **Wash your hands** (and wear gloves if you have them).
- **Irrigate with lots of clean water** — tap water you'd drink is fine; if not, use **boiled and cooled water**. Pour a steady stream, or squirt it with a clean plastic bag with a pin-hole or a large syringe, to wash out dirt. More water = better.
- Wash around the wound with **soap and water**. Pick out dirt and gravel with **tweezers cleaned with alcohol**.
- **Don't pour hydrogen peroxide, iodine, or rubbing alcohol into the wound** over and over — they damage healing tissue. Plain water and soap work. *(Red Cross; Where There Is No Doctor)*

3. Cover it

- A thin layer of **petroleum jelly or antibiotic ointment** (or honey — below), then a clean bandage.
- **Change the dressing daily**, or when wet or dirty, and look at the wound each time.

4. Close it — or leave it open?

- **Small, clean, straight cuts** with edges that come together easily can be closed with **butterfly strips or adhesive closure strips** after cleaning.
- **Leave OPEN** (cover with a clean, moist dressing and let it heal from the bottom up): **animal bites, punctures, dirty or crushed wounds, and anything older than about 12 hours.** Closing these traps germs inside. *(Where There Is No Doctor)*
- **Deep, gaping wounds** need a doctor for stitches or glue — ideally **within hours**.
- **Don't** use super glue from the hardware store on deep or dirty wounds. (Medical skin glue is different.)

5. Watch for infection (check every day)

Normal: a little pink at the edge, mild soreness, clear fluid the first day or two.

Infection signs: increasing **redness, swelling, warmth, or pain** after day 2; **pus**; **red streaks** spreading up the arm or leg; **fever**; foul smell. → **Get medical care.** Spreading redness or red streaks means the infection is moving — that's urgent.

Honey on wounds and burns

Honey has been used on wounds for thousands of years, and modern studies back it up for some uses: a Cochrane review found **honey dressings heal partial-thickness (blistered) burns faster than some conventional dressings**. Evidence for other wounds is mixed. *(Cochrane Review, Jull et al.)*

- **Best: medical-grade honey** (sterilized, sold as wound gel or dressings, e.g. "Medihoney"). Keep a tube in the kit.
- **Kitchen or raw honey** is not sterile and may contain bacterial spores. As a last resort on adults' wounds, only after thorough cleaning.
- **Never use raw/kitchen honey on an infant's wound**, and **NEVER give honey by mouth to a baby under 1 year** (botulism risk). *(CDC, AAP)*
- Spread a thin layer on the dressing (not directly into a deep wound), cover, change daily.

Splinters

- Wash. Use **tweezers cleaned with alcohol**; pull out in the same direction it went in.
- For a splinter under the skin: sterilize a needle (alcohol or flame, then cool), gently lift the skin over the end, then tweeze.
- Wash again and cover. **Deep splinters, wood under a fingernail, or any you can't remove → doctor** (wood left in skin often gets infected).
- **Cactus spines** (prickly pear "glochids" — the tiny hair-like ones): press **duct tape** or a layer of white glue over the area, let dry, and peel off.

Tetanus — a real farm risk

Tetanus ("lockjaw") lives in **soil, dust, and manure** — including **chicken and rabbit manure, barnyards, and rusty barbed wire**. It enters through **punctures, deep cuts, nail wounds, burns, and animal scratches and bites.** It's rare because of vaccination, but often deadly when it happens.

- **Adults need a tetanus booster (Td or Tdap) every 10 years.** For a **dirty or deep wound**, a booster is recommended if it's been **more than 5 years** since the last one. *(CDC)*
- **Kids** get tetanus protection with their regular DTaP shots — keep them on schedule.
- **Do it now:** check **everyone's** last tetanus date, including Grandma and Grandpa, and write it in the Home Worksheet. Mom can get Tdap while breastfeeding.

Animal scratches and bites on the farm

- **Rabbit scratches** (from kicking hind legs) and **chicken pecks**: wash right away with soap and running water for several minutes; cover; watch for infection.
- **Cat and dog bites** infect easily — wash well and **see a doctor** for antibiotics, especially bites to the hand.
- **Wild animal bites** or a **bat found in a room where someone slept** (especially a child) → call the doctor or county health department the same day about **rabies shots**. Rabies is fatal once symptoms start, and skunks and bats in Colorado carry it.

Burns

1. **Cool running water for 20 minutes** — the single best burn treatment. Cool, not ice-cold. Remove rings, watches, and tight clothes. *(Red Cross / American Burn Association)*
2. **Never** put **ice, butter, oil, toothpaste, or flour** on a burn.
3. **Don't pop blisters.**
4. Cover loosely with **clean plastic wrap** or a non-stick dressing. **Honey** (medical-grade) or **aloe vera gel** can soothe and help minor burns heal. *(Some evidence for both.)*
5. Pain medicine by weight (see below).
6. **ER:** see "Get help NOW." Any burn on a baby or toddler beyond a tiny one should be seen.

Wood stove and hot liquid burns are the most common burns in young kids — gates and "hot!" rules prevent most of them.

Fevers — what fever is and what to do

Fever is the body **fighting infection**. It is usually **not dangerous by itself** in children over 3 months; how the child **acts** matters more than the number. The goals are to keep them **comfortable and well hydrated**, and to **spot the dangerous illnesses** early. *(AAP)*

At home

1. **Fluids, fluids, fluids.** Breast milk, formula, water, broth, popsicles, oral rehydration solution. Fever and dry Colorado air both dry kids out fast.
2. **Dress lightly**; one light blanket. Don't bundle a feverish child.
3. **Lukewarm sponge bath** if they're uncomfortable — **lukewarm, not cold**. **Never use cold water, ice, or rubbing alcohol** (alcohol can be absorbed and poison a child).
4. **Rest.** Keep sick people away from the babies as much as possible; wash hands.
5. **Medicine is for comfort, not to reach a "normal" number.** Use only the doses below.

Oral rehydration solution (ORS)

Store-bought ORS (Pedialyte or similar, or ORS powder packets) is best — stock some. If you run out, the **WHO / Rehydration Project home recipe** is:

> **1 liter (about 1 quart) of clean, boiled-and-cooled water + 6 level teaspoons of sugar + ½ level teaspoon of salt.** Stir until dissolved. Use within 24 hours.

- **Measure carefully** — too much salt is dangerous for children; too much sugar worsens diarrhea. A little too dilute is harmless.
- Give **small sips often** (a spoonful every minute or two if vomiting).
- Roughly: under 2 years, ¼–½ cup after each watery stool; 2 and older, ½–1 cup. *(WHO / Rehydration Project)*
- **Babies under 6 months:** keep breastfeeding/formula; don't give homemade ORS unless a doctor says to.
- Add a little orange juice or mashed banana for potassium if available.

Fever and pain medicine — doses by WEIGHT

From the AAP (HealthyChildren.org) dosing tables. ***Use weight, not age, when you know it.*** *Use the syringe that comes with the medicine, never a kitchen spoon. Check the strength on the box — it must match the table.*

Acetaminophen (Tylenol) — liquid **160 mg per 5 mL** (infants' and children's liquids are the same strength). Every **4–6 hours** as needed; **no more than 5 doses in 24 hours**.

Weight	Usual age	Dose
6–11 lb	0–3 months	**Doctor only** — fever at this age = go to care
12–17 lb	4–11 months	2.5 mL (80 mg)
18–23 lb	12–23 months	3.75 mL (120 mg)
24–35 lb	2–3 years	5 mL (160 mg)
36–47 lb	4–5 years	7.5 mL (240 mg)
48–59 lb	6–8 years	10 mL (320 mg)

Ibuprofen (Motrin, Advil) — NOT under 6 months old. Every **6–8 hours** as needed; **no more than 4 doses in 24 hours**. Give with food. Avoid if the child is vomiting a lot or dehydrated.

Weight	Usual age	Infant drops (50 mg / 1.25 mL)	Children's liquid (100 mg / 5 mL)
under 12 lb	under 6 months	**Do not use**	**Do not use**
12–17 lb	6–11 months	1.25 mL	2.5 mL
18–23 lb	12–23 months	1.875 mL	3.75 mL
24–35 lb	2–3 years	—	5 mL
36–47 lb	4–5 years	—	7.5 mL
48–59 lb	6–8 years	—	10 mL

Infant ibuprofen drops are twice as strong as children's liquid — don't mix them up.

Rules:

- **NEVER give aspirin (or willow bark, or Pepto-Bismol, which contains a salicylate) to children or teens** — it can cause **Reye's syndrome**, a deadly liver and brain illness. *(AAP, FDA)*
- **Don't give two products that both contain acetaminophen** (many cold medicines do). Overdose damages the liver.
- **No cough and cold medicines under age 4**; check with the doctor under 6. *(FDA, AAP)* For coughs in kids **over 1 year**, a **spoonful of honey** helps as much as many cough medicines. *(AAP)*
- **Write down every dose and time** (use the log page).
- **Breastfeeding Mom:** acetaminophen and ibuprofen at normal adult doses are generally considered compatible with breastfeeding *(LactMed)* — confirm with her doctor.

Herbs and kitchen remedies — what the evidence says

The plains around Wiggins and our garden and kitchen have some traditional remedies. Here is an honest rating. **Strong** = good human studies; **Some** = a few studies or a well-documented traditional use; **Weak** = mostly tradition or lab studies.

Safety first:

- **Identify plants with 100% certainty** — learn them from CSU Extension or an experienced local in summer. Deadly look-alikes grow by the South Platte (water hemlock, poison hemlock — Chapter 7).
- **Pregnant or breastfeeding Mom and the babies: avoid herbal medicines** except normal food amounts (like ginger or garlic in cooking, or weak chamomile tea for Mom), unless a doctor okays them. Little is known about their safety for infants. *(NCCIH)*
- Allergies: **yarrow, chamomile, and echinacea** are in the **ragweed/daisy family** — avoid if allergic to ragweed.
- Herbs can interact with medicines (especially blood thinners, which Grandma or Grandpa may take).

Remedy	Where	Traditional use	Evidence	Cautions
Honey	kitchen / beekeepers	wounds, burns, cough	**Strong** (burns, cough in kids over 1)	**Never by mouth under 1 year**; medical-grade for wounds
Aloe vera gel	houseplant	minor burns, sunburn	**Some**	skin use only; patch test
Ginger (tea, fresh)	kitchen	nausea, stomach upset	**Some–Strong** for nausea	food amounts fine; ask doctor about large doses
Chamomile tea	garden	upset stomach, sleep, calm	**Weak–Some**	ragweed allergy; **not for babies**
Yarrow (*Achillea millefolium*)	native — roadsides, pastures	packed on small cuts to slow bleeding	**Weak** (traditional only; no clinical trials) *(EMA)*	ragweed-family allergy; not in pregnancy; small, superficial wounds only
Plantain (*Plantago major*, broadleaf weed)	yards, farmyards	crushed leaf on bites, stings, scrapes	**Weak**	clean leaves; still wash wound first
Prickly pear pad gel	dry pastures	burns, scrapes (pad split open)	**Weak**	remove every spine and glochid; clean first
Garlic	garden	eaten for colds	**Weak**	**NEVER put raw garlic on skin** — causes chemical burns, especially on kids
Sage (garden sage)	garden	gargle for sore throat	**Weak–Some**	not concentrated oil; avoid large amounts in pregnancy
Elderberry (cooked syrup)	river bottoms / shrubs	colds and flu symptoms	**Weak–Some** (small studies) *(NCCIH)*	**raw berries, leaves, stems are poisonous** — cook ripe berries only; safety unknown for pregnancy, breastfeeding, babies
Salt-water gargle	kitchen	sore throat	**Some**	½ tsp salt in a cup of warm water; kids old enough to gargle

Remedy	Where	Traditional use	Evidence	Cautions
Willow bark	creeks	pain, fever	works like aspirin	**NEVER for children** (Reye's syndrome)

Bottom line: keep **honey, ORS, acetaminophen, ibuprofen, a thermometer, and clean water** stocked — they beat any herb.

Dangers specific to our area

Rattlesnake bites

Prairie rattlesnakes live in the grasslands, rocky areas, and outbuildings of eastern Colorado, mostly April–October. *(Colorado Parks and Wildlife)*

Do:

- **Get away from the snake.** Call 911 / **drive to the ER now** — antivenom is the only real treatment.
- Keep the person **calm and still**; keep the bitten limb **still, at about heart level**.
- **Remove rings, watches, and tight clothes** before swelling starts.
- Note the time; mark the edge of swelling with a pen every 15 minutes.

Do NOT: cut the bite, **suck out venom**, use a **tourniquet**, put **ice** on it, give **alcohol**, use electric shock, or try to catch or kill the snake. A dead snake can still bite by reflex. *(CDC, Red Cross)*

Prevention: boots and long pants in tall grass; look before reaching into woodpiles, hay stacks, and under coops and hutches; keep grass mowed around buildings; teach kids to **freeze and back away** from any snake.

Heat illness

Summers here are hot, dry, and sunny at 4,500 ft — sweat evaporates fast, so you may not notice how much you're losing.

- **Heat exhaustion:** heavy sweating, weakness, dizziness, headache, nausea, cool clammy skin → move to shade, loosen clothes, cool wet cloths, sip water or ORS.
- **Heat stroke: hot skin, confusion, fainting, may stop sweating → 911.** Cool them fast (wet them down, fan, ice packs to neck, armpits, groin) while waiting. *(CDC)*
- Babies and Grandma and Grandpa overheat fastest. **Never leave anyone in a vehicle.**

Cold — frostbite and hypothermia

See Chapter 8. Wind on the open plains makes it worse. **Warm-water rewarming, no rubbing, no fire** for frostbite; **warm the core first** for hypothermia.

Dehydration — dry air and altitude

Our air is **very dry** and we're at about **4,500 feet**; you lose water just breathing, especially working outside or by the wood stove in winter. **Drink before you're thirsty.** Check urine: pale yellow is good, dark is a warning. Nursing Mom needs extra. Kids: see Chapter 4.

Hantavirus — from mouse dust

Hantavirus is a rare but **often deadly** lung infection carried by **deer mice**, common in barns, sheds, coops, feed rooms, and vehicles on the plains. People catch it by **breathing dust** from mouse droppings, urine, and nests. Colorado has cases most years. *(CDC; Colorado Dept. of Public Health and Environment)*

Safe cleanup (CDC):

1. **Air it out:** open doors and windows **30 minutes** before cleaning; leave while it airs.
2. **Gloves on. Do NOT sweep, vacuum, or use a leaf blower** — that throws virus into the air.
3. **Spray droppings and nests until soaking wet** with disinfectant or **bleach solution: 1½ cups bleach per gallon of water** (made fresh). **Let soak 5 minutes.**
4. Wipe up with paper towels; bag and toss. Mop the area with the bleach solution.
5. Wash gloved hands, then remove gloves and wash hands.

Symptoms (1–8 weeks after exposure): fever, muscle aches (thighs, hips, back), fatigue, then **coughing and shortness of breath**. If this happens after cleaning a barn or shed, **see a doctor right away and tell them about the mice.**

Prevention: store feed in **metal cans with lids**; seal gaps in the house; keep the coop and rabbit areas free of spilled feed.

Tularemia — from rabbits

Tularemia ("rabbit fever") is a bacterial infection carried by **rabbits, rodents, and ticks and deer flies**; it shows up in Colorado most years. People catch it by **handling or butchering infected rabbits**, tick and deer fly bites, or **breathing dust** (e.g., mowing over a dead rabbit). *(CDC; Colorado DPHE)*

- **Wear gloves when butchering** rabbits (Chapter 6). Don't butcher a rabbit that seemed sick or sluggish. **Never handle wild rabbits found dead** with bare hands.
- **Cook rabbit thoroughly** (160°F).
- Use **insect repellent** and do **tick checks**.
- **Symptoms:** sudden fever, chills, swollen lymph nodes, a skin ulcer where bacteria entered. **See a doctor and mention rabbits** — it's treatable with antibiotics.

Also in our area

- **West Nile virus:** mosquitoes near the South Platte and irrigation ditches. Use **repellent (DEET, picaridin)** at dawn and dusk; dump standing water. Most people get mild illness, but older adults are at highest risk of serious disease. *(CDPHE)*
- **Plague:** carried by fleas on **prairie dogs** and other rodents in Colorado. Keep pets away from prairie dog colonies; use flea control; don't handle dead rodents. Sudden high fever with a painful swollen lymph node → doctor now.
- **Salmonella from chickens:** wash hands after handling birds or eggs (Chapter 5).

Home remedy kit (add to the first aid kit)

- [ ] Medical-grade honey (wound gel) + a jar of regular honey (cough, ages 1+)
- [ ] ORS packets or Pedialyte; sugar and salt with the recipe taped to the jar
- [ ] Infants' and children's acetaminophen; infants' and children's ibuprofen; oral syringes; this dose table
- [ ] Digital thermometer (rectal for the baby)
- [ ] Butterfly closures / wound closure strips; non-stick dressings; petroleum jelly
- [ ] Irrigation syringe (large, 20–60 mL); clean water stored
- [ ] Tweezers, needles, magnifier, duct tape (cactus spines)
- [ ] Commercial tourniquet; gloves
- [ ] Aloe vera gel; ginger; chamomile tea
- [ ] Bleach + spray bottle (hantavirus cleanup); insect repellent; tick remover
- [ ] Everyone's tetanus dates written in the Home Worksheet

CHAPTER 10

Power, Light & Communication

Power

Generators — safe use *(CDC, FEMA)*

- **Outdoors only, at least 20 feet from windows, doors, and vents.** Never in the house, basement, garage, or barn with the door open.
- **Never backfeed** through a house outlet. Use a **transfer switch or interlock** installed by a licensed electrician, or plug appliances directly into the generator with heavy outdoor extension cords.
- Let it cool before refueling. Store fuel in approved cans, away from the house; add stabilizer and rotate.
- **Priority loads:** furnace blower/controls, well pump, freezer/fridge (run a few hours at a time), lights, phone charging, coop waterer. Check each one's watts.
- An **inverter generator** is quieter and safer for electronics.

Without a generator

- **Portable power stations** (battery boxes) + **solar panels** charge phones, lights, radios, a breast pump, and small fans.
- **Car** = phone charger (run it outside, never in a closed garage; clear snow from the tailpipe).
- **Batteries:** AA/AAA stock for radios and lights; rechargeables + solar charger.

Light

- **LED headlamps** for every adult (hands-free for diapers and chores) and LED lanterns for rooms.
- **Avoid candles and oil lamps** around toddlers — a leading cause of house fires in outages. If used, never leave burning unattended.
- Glow sticks for kids' nightlights.

Communication and information

Getting warnings

- **NOAA Weather Radio** with battery and hand-crank backup and **SAME** alerts set for **Morgan County**. Our forecasts come from the **National Weather Service office in Boulder** (Denver/Boulder). Write our local transmitter frequency in the Home Worksheet.

- **AM radio: KOA 850 AM (Denver)** is Colorado's main emergency broadcast station and carries far at night.
- Phones: turn on **Wireless Emergency Alerts**; sign up for Morgan County alerts.

Talking to each other

Option	License?	Use for
FRS walkie-talkies	No license	Between our house and Grandma & Grandpa's, the barn, the coop. Short range.
GMRS radios	**FCC license — no test**, one license covers the family	Longer range than FRS, can use repeaters.
Amateur (ham) radio	**Exam required** (Technician license is the entry level)	Talk across the county or state; local ham clubs often help in disasters.
CB radio	No license	Truckers and farm neighbors; short-medium range.

- Agree on **check-in times** (e.g., on the hour) and a **family channel**.
- **Out-of-area contact:** long-distance texts often work when local calls don't.
- **Texts beat calls** when networks are jammed.

Do it now

- [ ] NOAA Weather Radio (battery + crank), set to Morgan County.
- [ ] Headlamps for adults, lanterns, batteries.
- [ ] FRS/GMRS radio set (get the GMRS license online from the FCC).
- [ ] Power station + solar panel, or a generator with a professionally installed inlet.
- [ ] Phone car chargers and power banks.

CHAPTER 11

Security, Neighbors & Working Together

Families survive hard times **together**. Having Grandma and Grandpa next door is a huge strength. Good neighbors are worth more than any supply.

Two houses, one team

- **Decide now:** if heat or water runs short, **which house do we consolidate into?** (Usually the one with the reconnected wood stove, water access, and room for babies.)
- **Divide the jobs:**

Job	Who (fill in)
Babies & kids	
Heat & firewood	
Water	
Chickens & rabbits	
Cooking & food inventory	
Radio, news & check-ins	
Night watch / fire & CO watch	
Medical & medicines	

- **Daily family meeting** (5 minutes): what we have, what we need, today's jobs, health check on everyone.
- Grandma and Grandpa's knowledge (cooking from scratch, gardening, preserving) is valuable — and their health needs (medicines, mobility, staying warm) come first too.

Neighbors

- **Know them before trouble.** Trade phone numbers; learn who has skills (nurse, mechanic, electrician, farmer, hunter) and needs (elderly, medical equipment, new babies).
- **Trade and share:** eggs, rabbit, kits, hay, firewood, and labor are excellent trade goods. A neighbor network can watch for fire, share a generator, or butcher together.
- Join or start a simple **neighborhood check-in** (phone tree or radio channel).

Staying safe

- **Keep a low profile.** Don't advertise how much food, fuel, or water we have. Stack supplies out of sight.

- **Lock up:** house, outbuildings, fuel, feed, and generator (chain it). Animal theft rises in hard times.
- **Motion lights** (solar) and a **dog** are good early warnings.
- **Night watch** in rotation during unrest — also watching for chimney fires and CO alarms.
- **Avoid confrontation.** Most problems are solved by staying home, staying calm, and calling the Sheriff.
- **Firearms:** only adults trained and lawful; **locked storage** with four kids in the home — always. Know Colorado law on self-defense and safe storage.
- **Scams and rumors:** check news from official sources (Sheriff, NWS, county) before acting.

Taking care of our heads

- Stress is normal. Keep routines, sleep in shifts, eat regularly, step outside for daylight.
- Talk to each other. Give the 6-year-old honest, simple answers.
- Watch for signs a family member is struggling (not sleeping or eating, withdrawing, hopelessness). Colorado Crisis Services and the **988** Suicide & Crisis Lifeline help by phone or text.

Do it now

- [ ] Sit down with Grandma and Grandpa: consolidation plan, job list, check-in plan.
- [ ] Neighbor phone list (Home Worksheet).
- [ ] Locks, solar motion lights, gun safe.

CHAPTER 12

Plains Weather & Disasters

Know the threats where we live and what to do in each.

Blizzards and extreme cold

- **Before:** fill propane, stock firewood, food, water, medicines; charge everything; bring animals' water and feed in close.
- **During: stay inside.** Plains blizzards cause **whiteouts** — people get lost between the house and barn. If you must go to the animals, **tie a rope line** from house to barn and go in pairs.
- **Never travel** in a blizzard warning unless it's a life-or-death emergency.
- **Car winter kit:** blankets/sleeping bags, hats, gloves, boots, water, snacks, shovel, ice scraper, sand/kitty litter, jumper cables, flashlight, phone charger, flares, and baby supplies when kids ride.
- **If stranded in a car:** stay with the car, run the engine ~10 minutes an hour for heat with the **tailpipe cleared of snow** and a window cracked, tie bright cloth to the antenna. *(NWS / Ready.gov)*

Lessons from the December 2006 Holiday Blizzards

- **Two blizzards hit eastern Colorado about a week apart** (Dec 20–22 and Dec 28–30). Drifts reached **10–15 feet**; roads closed and homes were cut off; the National Guard and Civil Air Patrol searched for stranded drivers and dropped hay. *(Colorado DHSEM after-action reports)*
- **Thousands of cattle died** (estimates of 8,000–15,000). Many ranchers **had hay in the field but couldn't reach it** through the drifts. In a 1997 storm, about 27,000–30,000 cattle died.
- **Lessons for us:** expect a **second storm**; don't use up supplies in the first. **Stage feed, hay, and water right next to the animals** before the storm, not across a field. Keep a shovel at every door, and a sled or wagon to haul feed and water.

Tornadoes

- **Watch** = conditions are right. **Warning** = take shelter **now**.
- Go to the **basement** (or an interior room on the lowest floor without windows). Get under something sturdy; cover heads — **bike helmets for kids** help.
- Keep **shoes, a flashlight, radio, and baby bag** in the shelter area.
- Vehicles and mobile homes are not safe.

Hail and lightning

- Hail can be big here — stay inside away from windows; protect gardens with hail cloth; park under cover.
- **"When thunder roars, go indoors."** Wait 30 minutes after the last thunder. *(NWS)*

Grass and wildfire

- On **Red Flag Warning** days (dry + windy), no outdoor burning, welding, or driving through dry grass.
- **Defensible space:** mow and clear dry grass and weeds around the house, coop, hutches, propane tank, and woodpile.
- **Evacuate early** if officials say go — wind-driven grass fires move very fast. Know two ways out.
- Plan how to move or release animals.

Flooding (South Platte and ditches)

- **Turn around, don't drown. 6 inches of moving water can knock an adult down; 12 inches can carry away most cars.** *(NWS)*
- Move to higher ground if water rises. Keep kids away from floodwater (sewage, chemicals).
- After flooding, **treat well water as contaminated** until tested (Chapter 2).

Extreme heat and drought

- See "Summer heat" in Chapter 3. Water for people and animals first. Shade for rabbits and chickens (rabbits die from heat quickly).

Do it now

- [ ] Car winter kits in every vehicle.
- [ ] Shelter spot in the basement stocked (helmets, shoes, light, radio, baby bag).
- [ ] Mow defensible space around buildings and propane tank.
- [ ] Rope for a house-to-barn line.

CHAPTER 13

Seasonal Checklists & Inventory Sheets

Print these pages extra. Use pencil.

Fall (September–October) — the most important season

- [ ] **Propane tank filled**; gauge recorded.
- [ ] **Chimney swept/inspected**; stove tested with CO alarms on.
- [ ] Firewood stacked, covered, dry.
- [ ] CO and smoke alarms tested; new batteries.
- [ ] Hay for the whole winter; layer feed and pellets stocked.
- [ ] Heated waterers working; backup crocks/buckets ready.
- [ ] Coop light on timer; coop and hutches wind-blocked.
- [ ] Winter car kits in vehicles.
- [ ] Rotate water storage; check formula, diaper sizes, medicines, baby clothes sizes.
- [ ] Squash, pumpkins, potatoes stored.
- [ ] Generator test run (outdoors) with fresh fuel.
- [ ] Review this book with Grandma and Grandpa.

Winter (November–March)

- [ ] Watch the forecast daily; fill water before storms.
- [ ] Record propane % weekly.
- [ ] Keep snow off the propane tank lid/regulator and away from furnace vents outside.
- [ ] Mealworm bin going; sprouting fodder for hens.
- [ ] Plan breeding so kits aren't born in deep cold.
- [ ] Order garden seeds.

Spring (April–May)

- [ ] Order/hatch chicks to replace old hens.
- [ ] Start seeds indoors; plant cool crops; potatoes after soil warms.
- [ ] Check stored food; rotate.
- [ ] Tornado shelter spot restocked.
- [ ] Rotate water storage (every 6 months).

Summer (June–August)

- [ ] Garden: water, mulch, hail protection.
- [ ] Black soldier fly bin running; dry extra larvae for winter.
- [ ] Shade and cool water for rabbits and hens.
- [ ] Mow defensible space (fire season).
- [ ] Buy firewood early (cheaper, drier).
- [ ] Start the mealworm colony for winter.

Go-bag (one per person, by the door)

- [ ] Water (1 gal/person/day for 3 days if carrying is possible) + filter
- [ ] 3 days of food that needs no cooking; kids' snacks
- [ ] Baby: diapers, wipes, ready-to-feed formula + nipples (for the 1-year-old if still on formula; small backup for the newborn), sleep sack, change of clothes, carrier/wrap
- [ ] Medicines + list; copies of IDs, insurance, birth certificates (in a zip bag); cash
- [ ] Warm clothes, hats, gloves, rain gear, sturdy shoes
- [ ] Flashlight/headlamp, batteries, radio, phone charger/power bank
- [ ] First aid kit, hand sanitizer, toilet paper
- [ ] This book; a paper map of the county; a whistle
- [ ] Comfort item for each kid

Inventory sheet — supplies

Item	Target	On hand	Date checked	Expires
Drinking water (gal)	125			
Unscented bleach	1 gal			
Infant formula — ready-to-feed (backup)				
1-yr-old formula/milk (oz or cans)				
UHT/powdered milk				
Diapers — size ___				
Diapers — size ___				
Wipes				
Prenatal vitamins				
Grains (lb)	150/month			
Beans (lb)	40/month			
Oil/lard (gal)	2/month			
Canned goods (cans)				
Salt / sugar (lb)				
Layer feed (lb)				
Oyster shell (lb)				
Rabbit pellets (lb)				
Hay (bales)				
Propane (%)	> 30%			
Firewood (cords)				
Generator fuel (gal)				
Batteries AA / AAA / D				
Prescription meds (days)	30+			
First aid kit checked	yes			

Daily log (during an emergency)

Date	Propane %	Firewood used	Water used (gal)	Eggs	Rabbits	Notes (health, weather, events)

CHAPTER 14

Sources & Where to Learn More

The figures in this book come from these organizations. Visit them (while the internet works) for the latest guidance, and print what's useful.

Emergency planning

- FEMA / **Ready.gov** — water storage (1 gallon per person per day; 2-week home supply), winter storms, tornadoes, floods, generator safety, go-bags. ready.gov
- **National Weather Service** (Denver/Boulder office) — forecasts, NOAA Weather Radio, winter car safety, flood ("Turn Around, Don't Drown") and lightning safety. weather.gov/bou
- **Morgan County** Office of Emergency Management / Sheriff — local alerts and shelters.

Water

- **CDC** — making water safe in an emergency (boiling 1 minute; bleach use). cdc.gov/healthywater
- **EPA** — emergency disinfection of drinking water (bleach drop chart). epa.gov
- **Colorado Division of Water Resources** — well permits and records; rain barrel law (2 barrels / 110 gallons).

Heat and fire safety

- **NFPA 211** — Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances (clearances: 36 in for unlisted stoves, 18 in for single-wall connector, unless listed/protected). nfpa.org
- **EPA Burn Wise** — safe wood burning, seasoned firewood (<20% moisture), EPA-certified stoves. epa.gov/burnwise
- **Chimney Safety Institute of America (CSIA)** — certified sweeps, inspection levels, liners. csia.org
- **CDC** — carbon monoxide poisoning prevention; hypothermia and frostbite. cdc.gov
- Propane energy content (about 91,500 BTU/gallon) — U.S. Energy Information Administration / propane industry standard.

Babies and kids

- **American Academy of Pediatrics (AAP) / HealthyChildren.org** — breastfeeding frequency, switching to whole milk at 12 months (16–24 oz/day), never dilute or make homemade formula, safe sleep, fever under 3 months, dressing infants.
- **CDC** — breastfeeding mothers' extra calories (~450–500/day), breast milk storage, infant formula preparation and safe water, Salmonella from backyard poultry.
- **FDA** — warnings against homemade infant formula.
- **LactMed** (National Library of Medicine) — medicines during breastfeeding.

Food, chickens, and rabbits

- **Colorado State University (CSU) Extension** — backyard poultry, rabbits, gardening on the plains, frost dates, food preservation, wild plants. extension.colostate.edu
- **USDA** — food safety during power outages (freezer 48/24 hours, fridge 4 hours), safe cooking temperatures (rabbit and game 160°F), nutrient data (eggs, rabbit), Dietary Guidelines calorie estimates. usda.gov, fooddata central

- **National Center for Home Food Preservation** — tested pressure-canning recipes for meat. nchfp.uga.edu
- **Colorado Department of Agriculture** — State Veterinarian, avian influenza reporting.

Hunting, fishing, foraging

- **Colorado Parks and Wildlife** — licenses, seasons, CWD testing. cpw.state.co.us

Home remedies, wounds, and fevers

- **AAP / HealthyChildren.org** — acetaminophen and ibuprofen dosing tables by weight (acetaminophen 160 mg/5 mL; ibuprofen not under 6 months), fever guidance, honey for cough over age 1, no cough/cold medicine for young children.
- **American Red Cross**, **American Heart Association**, **Stop the Bleed**, **American Burn Association** — bleeding control, wound care, burn cooling (20 minutes).
- **CDC** — tetanus boosters (every 10 years; after 5 years for dirty wounds), hantavirus cleanup (ventilate 30 min, no sweeping, 1½ cups bleach/gallon, soak 5 min), tularemia, plague, West Nile, rabies, heat illness, snakebite first aid, infant botulism and honey.
- **Colorado Department of Public Health and Environment (CDPHE)** — hantavirus, tularemia, plague, rabies, and West Nile in Colorado. **Colorado Parks and Wildlife** — prairie rattlesnakes.
- **WHO / Rehydration Project** (rehydrate.org) — home oral rehydration recipe (1 L water, 6 level tsp sugar, ½ level tsp salt) and amounts after each stool.
- ***Where There Is No Doctor*** (David Werner, **Hesperian Health Guides**, free online) — wound cleaning and when to leave wounds open; a classic for care far from doctors. Its companion, ***Where There Is No Dentist***, is also worth printing.
- **NIH National Center for Complementary and Integrative Health (NCCIH)** — elderberry (preliminary evidence; raw parts poisonous), herbs for colds and flu, herb safety in pregnancy/breastfeeding.
- **European Medicines Agency (HMPC) monograph on yarrow** — traditional use only for small superficial wounds; ragweed-family allergy.
- **Cochrane Review: Honey as a topical treatment for wounds** (Jull et al.) — honey heals partial-thickness burns faster than some conventional dressings.
- **LactMed** (NIH) — medicines while breastfeeding. **Poison Control** 1-800-222-1222.
- **CommonSpirit St. Elizabeth Hospital, Fort Morgan** — ER location and phone (hospital website, October 2026).
- **CSU Extension** — identifying local plants (yarrow, plantain, prickly pear, elderberry) and poisonous look-alikes.

Experienced preppers, homesteaders, and survival guides

- **U.S. Army Survival Manual FM 21-76** (now **ATP 3-50.21**) — the "SURVIVAL" mindset and priorities.
- **The Church of Jesus Christ of Latter-day Saints** — "All Is Safely Gathered In," Longer-Term Food Supply, and the one-year basics list (400 lb grains, 60 lb legumes, etc. per adult; child percentages); **BYU** Department of Nutrition, Dietetics & Food Science shelf-life research. churchofjesuschrist.org/topics/food-storage
- **The Prepared** (theprepared.com) — winter survival and lessons from the 2021 Texas blackout (tent-in-a-room, body heat, diesel gelling, indoor-rated heaters only).

- **Joel Salatin / Polyface Farm** — the Raken (rabbit-over-chicken) house and deep-bedding method (8–12+ inches of carbon). Mother Earth News; McMurray Hatchery blog.
- **Colorado Division of Homeland Security & Emergency Management** — after-action reports on the December 2006 Holiday Blizzards; **Natural Hazards Center (CU Boulder)** Quick Response Report 191 on livestock losses.
- **Recommended books for the shelf** (classic, widely respected references): *SAS Survival Handbook* (John "Lofty" Wiseman); *When Technology Fails* (Matthew Stein) — off-grid water, heat, and food; *Storey's Guide to Raising Chickens* (Gail Damerow); *Storey's Guide to Raising Rabbits* (Bob Bennett); *The Encyclopedia of Country Living* (Carla Emery).

Medical training

- **American Red Cross** and **American Heart Association** — First Aid, CPR (adult/child/infant). **Stop the Bleed** — bleeding control. 988 Suicide & Crisis Lifeline. Poison Control 1-800-222-1222.

This handbook was written for our family in October 2026. It is general guidance and not a substitute for professional medical, electrical, chimney, or legal advice. Check guidance yearly — recommendations change.